

Program 8 :

Manifest Files :

```
abhi@abhi-ThinkPad-T490:~$ cd devops/program8
abhi@abhi-ThinkPad-T490:~/devops/program8$ cat nodejs-con.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-config
data:
  # A tiny Node.js HTTP server that uses APP_PORT and APP_MESSAGE from env
  server.js: |
    const http = require('http');
    const port = process.env.APP_PORT || 3000;
    const msg = process.env.APP_MESSAGE || 'Hello from Node.js';
    http.createServer((req, res) => {
      res.writeHead(200, {'Content-Type': 'text/plain'});
      res.end(msg + '\n');
    }).listen(port, () => {
      console.log(`Server listening on port ${port}`);
    });
  # Default environment values (strings)
  APP_PORT: "3000"
  APP_MESSAGE: "Server is up and Running.....!"

---
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      # Run the server file from the mounted ConfigMap
      command: ["node", "/app/server.js"]
      ports:
        - containerPort: 3000
      envFrom:
        - configMapRef:
            name: nodejs-config # injects APP_PORT and APP_MESSAGE as env vars
      volumeMounts:
        - name: config-volume
          mountPath: /app
          readOnly: true
  volumes:
    - name: config-volume
```

Step 2 : Kubernetes Apply and Deployments services running logs

```

abhi@abhi-ThinkPad-T490:~/devops/program8$ kubectl get deploymetns
error: the server doesn't have a resource type "deploymetns"
abhi@abhi-ThinkPad-T490:~/devops/program8$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2           11d
abhi@abhi-ThinkPad-T490:~/devops/program8$ kubectl apply -f nodejs-con.yaml
configmap/nodejs-config unchanged
pod/nodejs-pod created
abhi@abhi-ThinkPad-T490:~/devops/program8$ kubectl get svc
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP          11d
nginx-deployment    NodePort    10.97.242.81 <none>        80:31551/TCP     140m
abhi@abhi-ThinkPad-T490:~/devops/program8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
my-cronjob-29425205-qmj84           0/1     Completed  0           12m
my-cronjob-29425210-ggnlv           0/1     Completed  0           7m35s
my-cronjob-29425215-ql5dk           0/1     Completed  0           2m35s
my-job-s8bpg                        0/1     Completed  0           10d
my-nginx                          1/1     Running    1 (10d ago)  10d
nginx-deployment-cdf8c8ff6-lcmsx    1/1     Running    1 (10d ago)  10d
nginx-deployment-cdf8c8ff6-xgthx    1/1     Running    1 (10d ago)  10d
nginx-pod                          1/1     Running    1 (10d ago)  10d
nodeapp-simple                     0/1     ImagePullBackOff  0           11d
nodejs-pod                         1/1     Running    0           23s
abhi@abhi-ThinkPad-T490:~/devops/program8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000

```

Browser verification :

